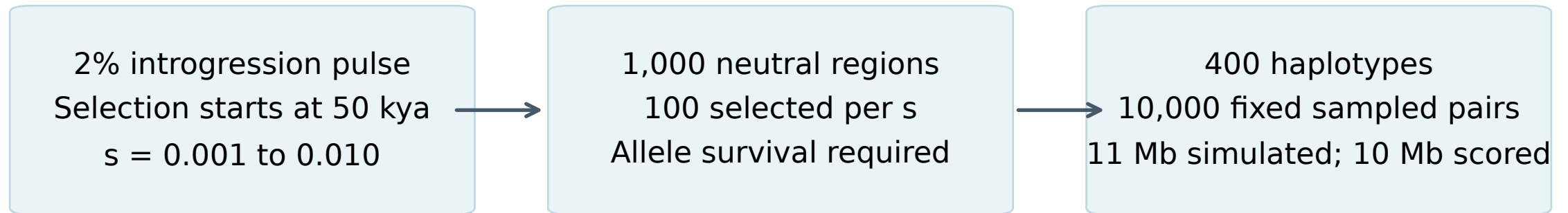


An archaic-carrier coalescence scan

Completed simulation design and the primary score



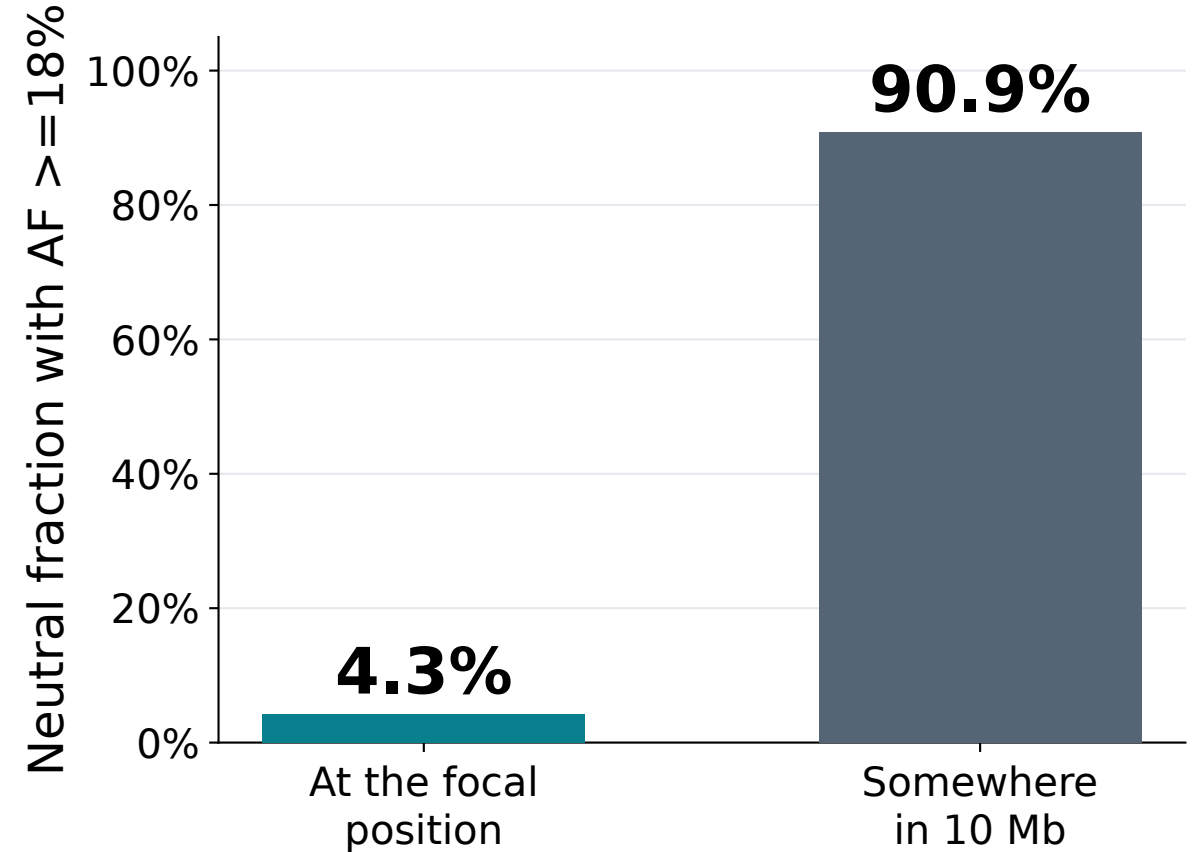
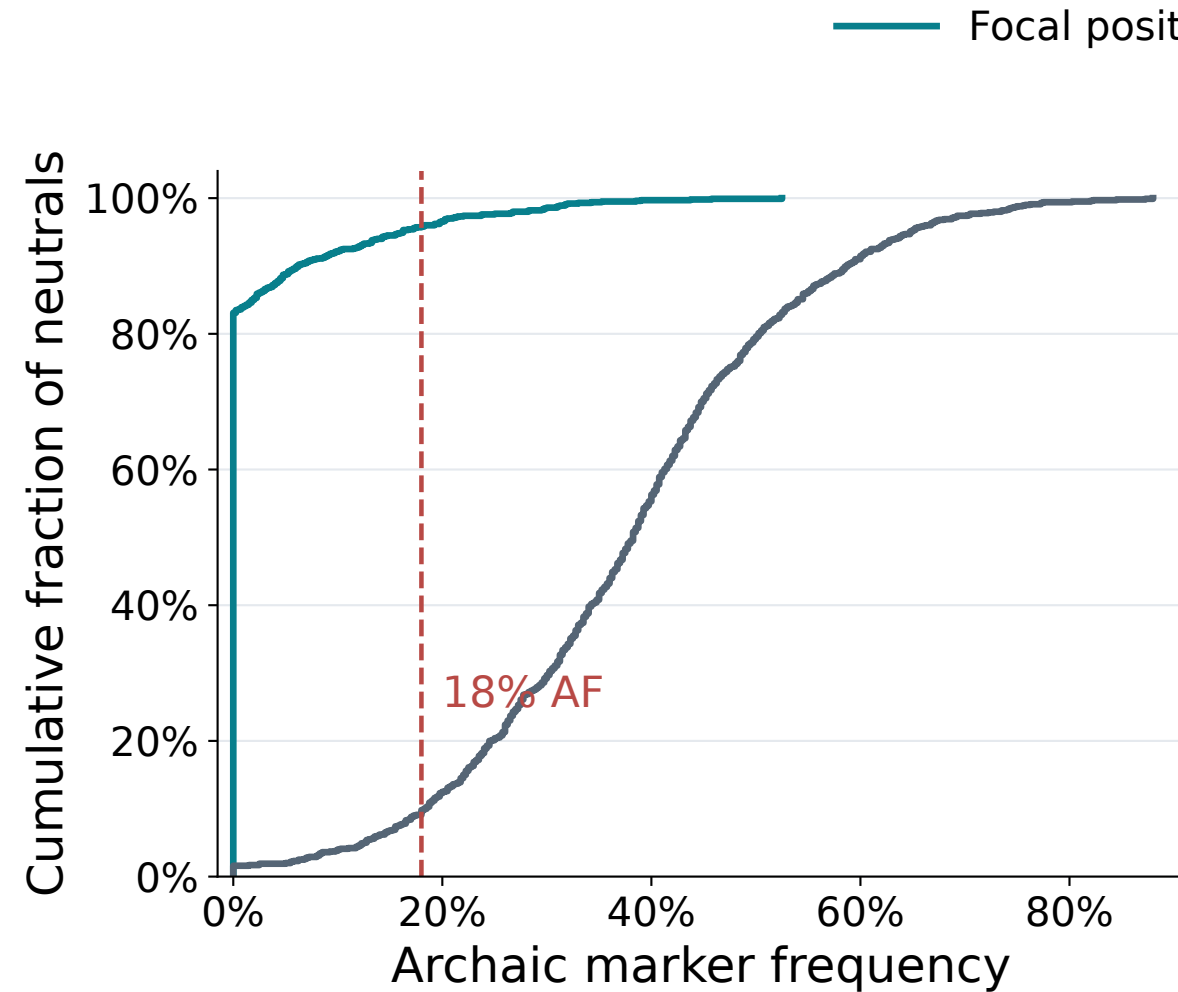
$$S_T = \text{frac_recent}_T(\text{ALT/ALT}) \times \frac{n_{\text{ALT/ALT}}}{n_{\text{all pairs}}} = \frac{n_{\text{recent ALT/ALT}, T}}{n_{\text{all pairs}}}$$

Eligible archaic ALT: oracle-specific origin after the archaic-modern split, before the pulse.

Truth uses local TMRCA; decoding uses posterior-mean TMRCA hard calls.

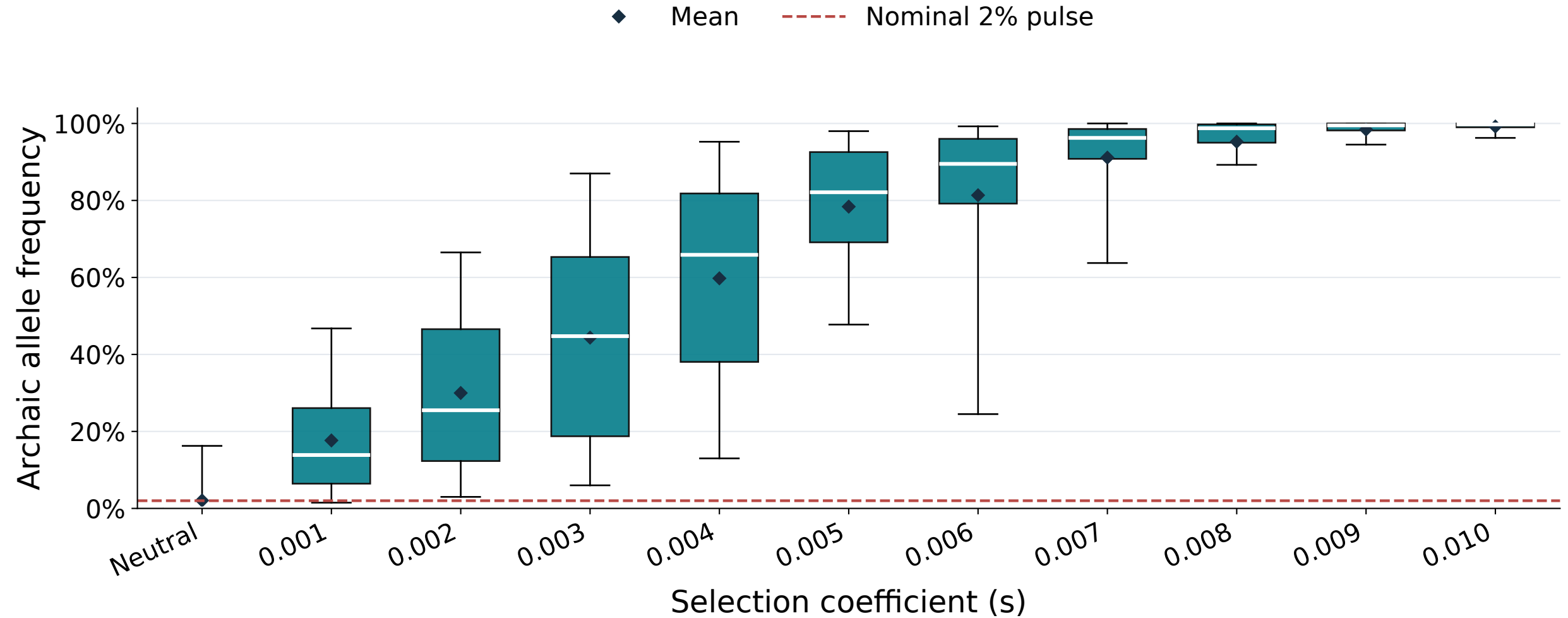
The null must match the position being tested

Same 1,000 neutral simulations; different spatial questions



Selection shifts local archaic-allele frequency

Completed cohort: 100 selected replicates per coefficient; 1,000 neutral focal positions

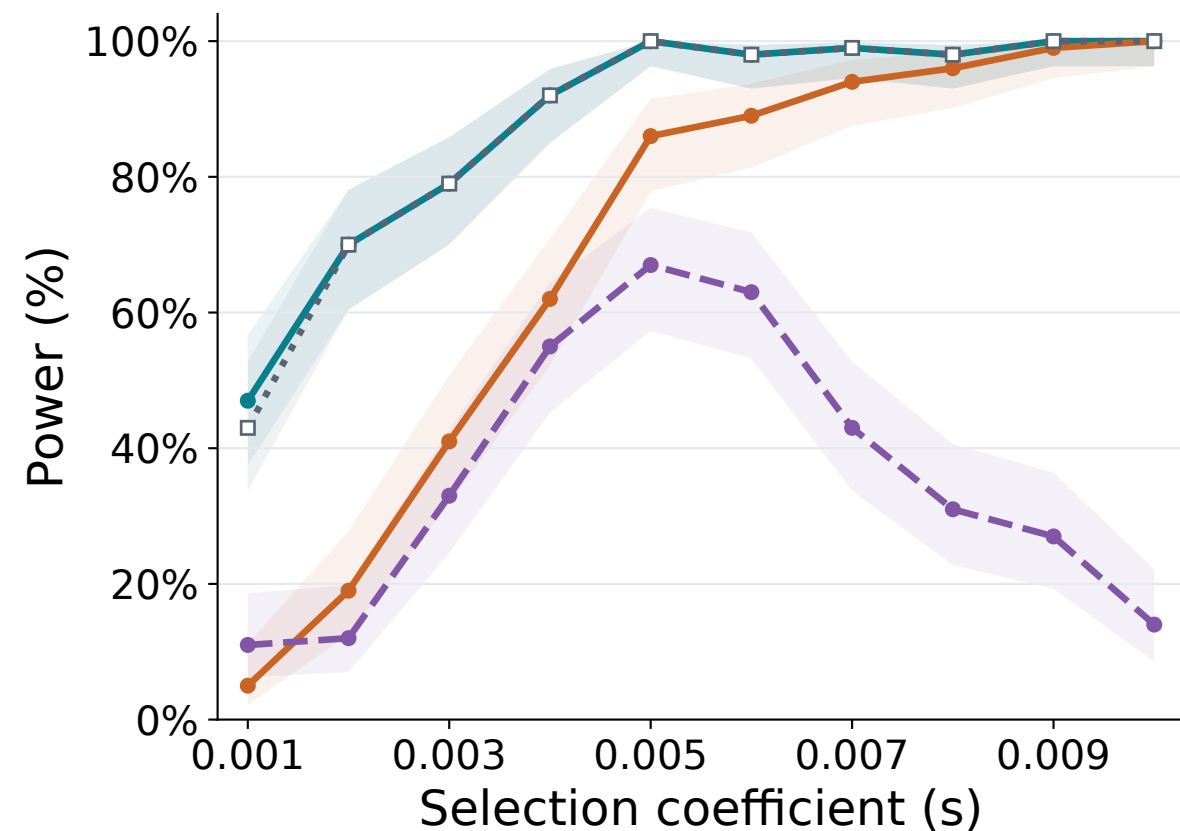


Local calibration recovers partial-sweep power

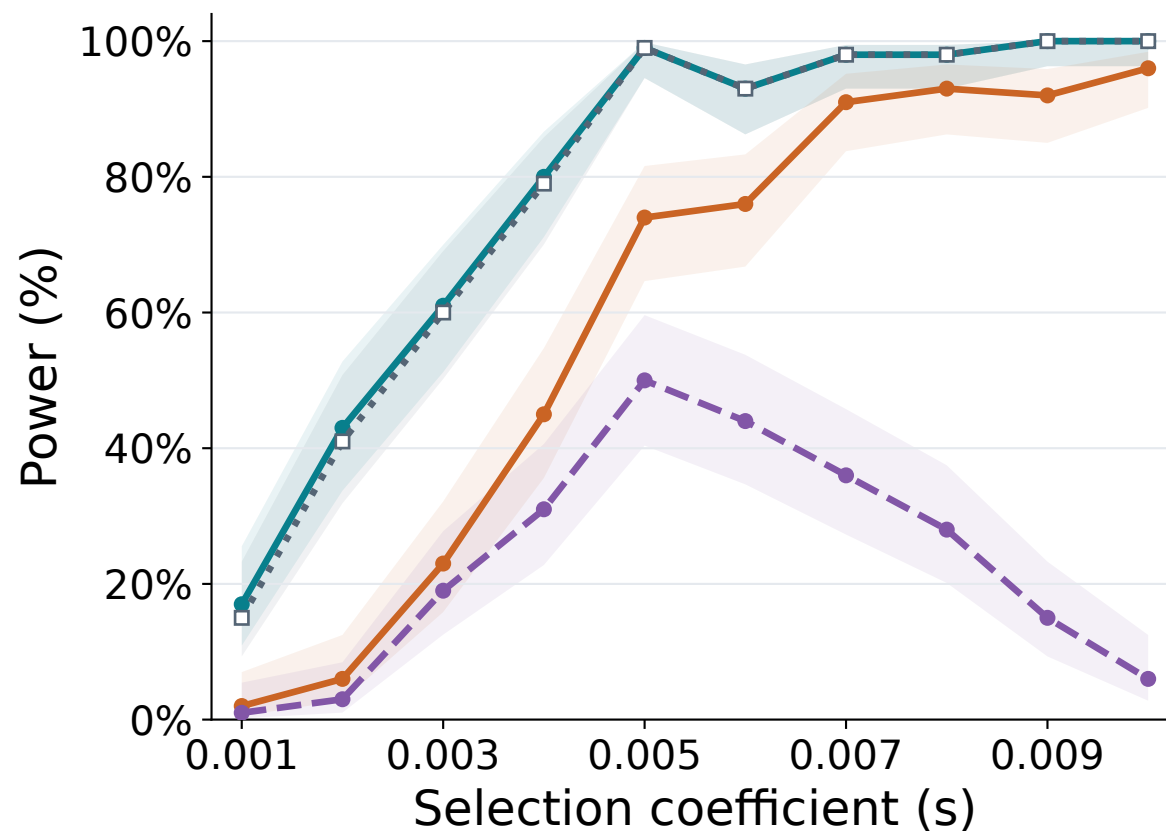
T = 50 kya for TMRCA scores | shaded bands: 95% Wilson intervals conditional on the fitted null

···□··· Archaic AF —●— ALT/ALT carrier mass —●— All-pair recency -●- Positional iHS

Nominal $p \leq 0.05$

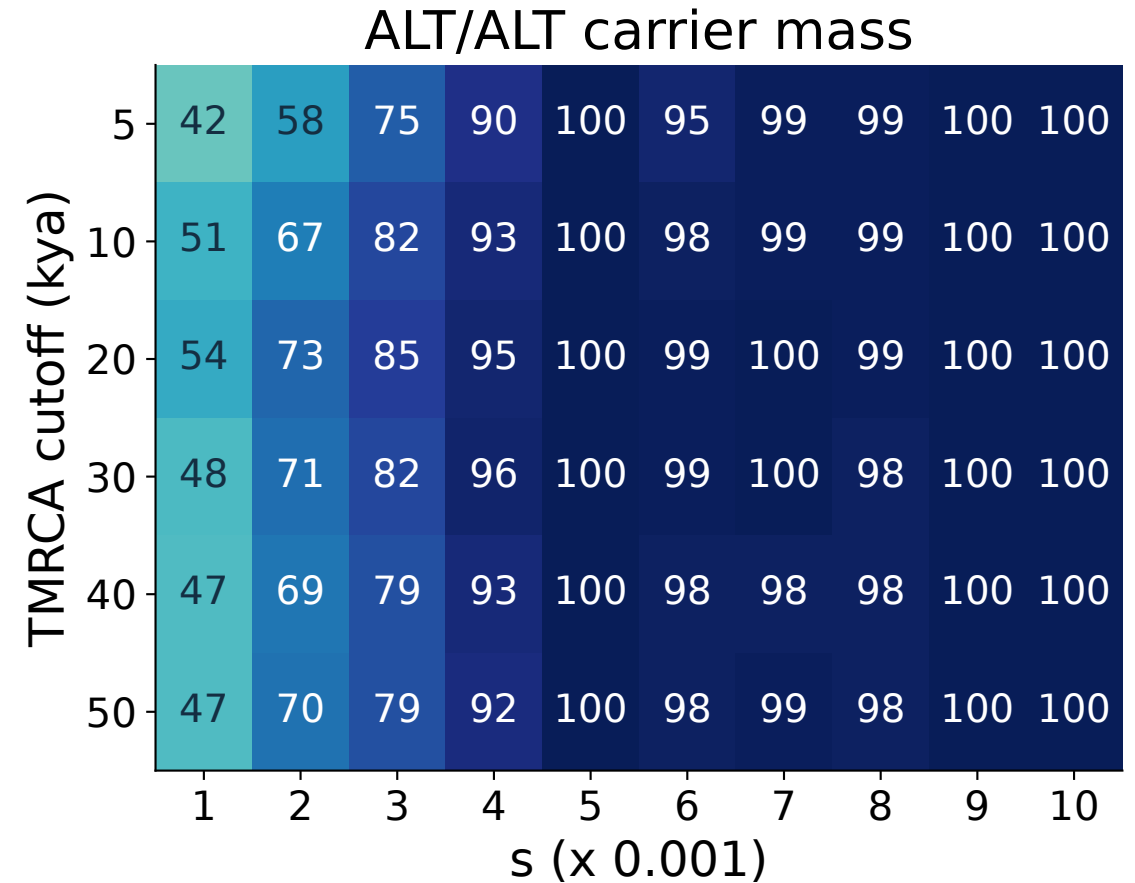
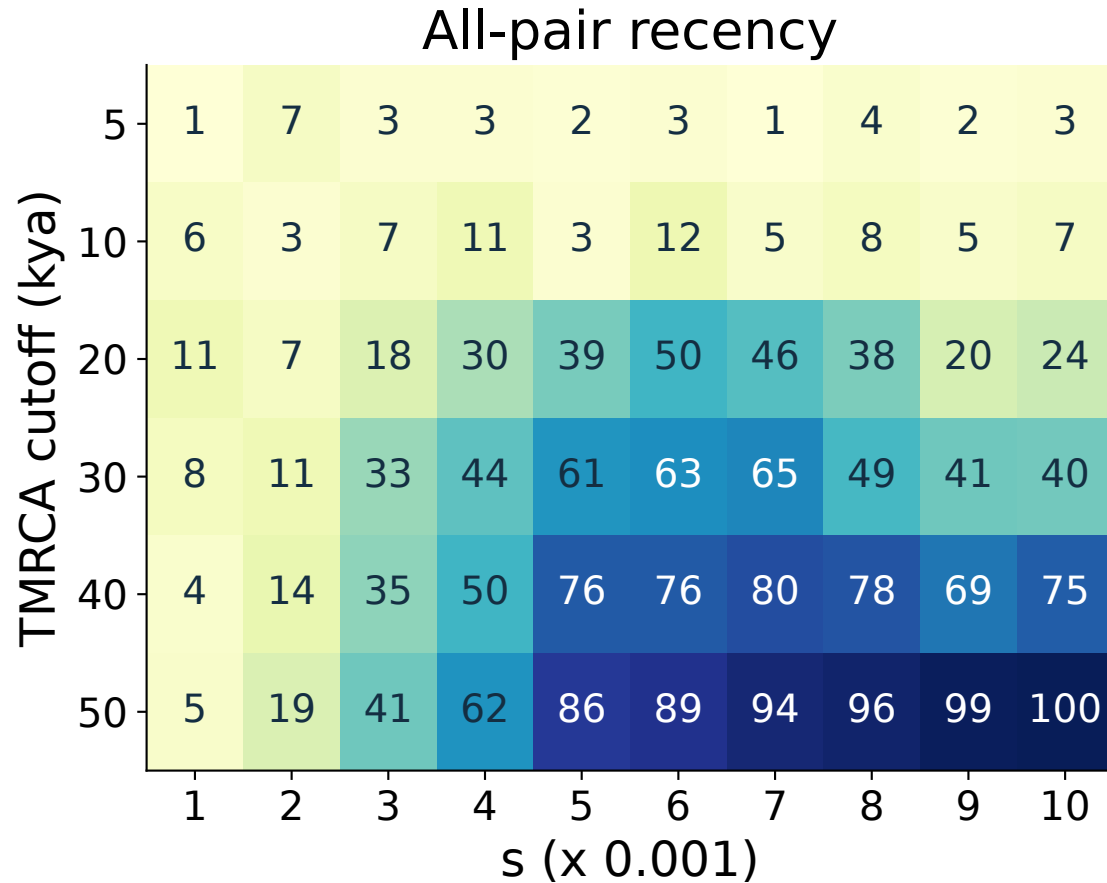


Nominal $p \leq 0.01$



Recency cutoffs change power

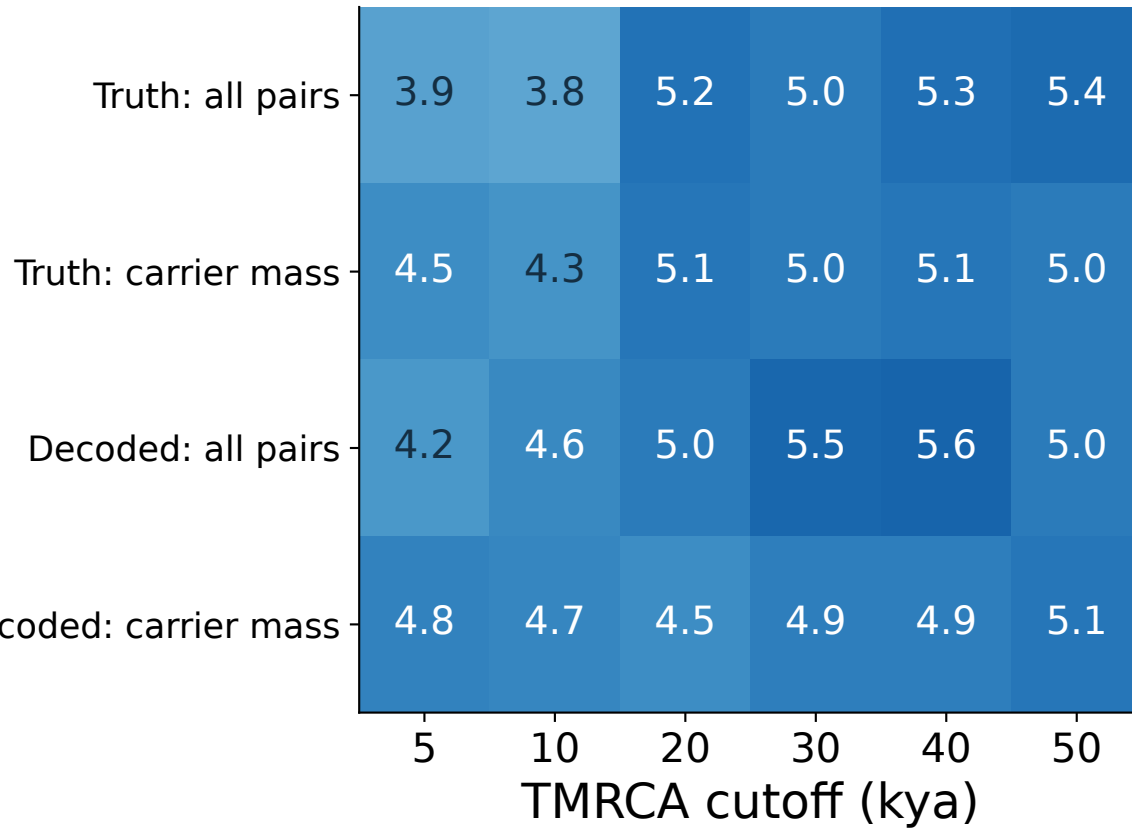
True TMRCA | $p \leq 0.05$ | every cell is a separately calibrated test



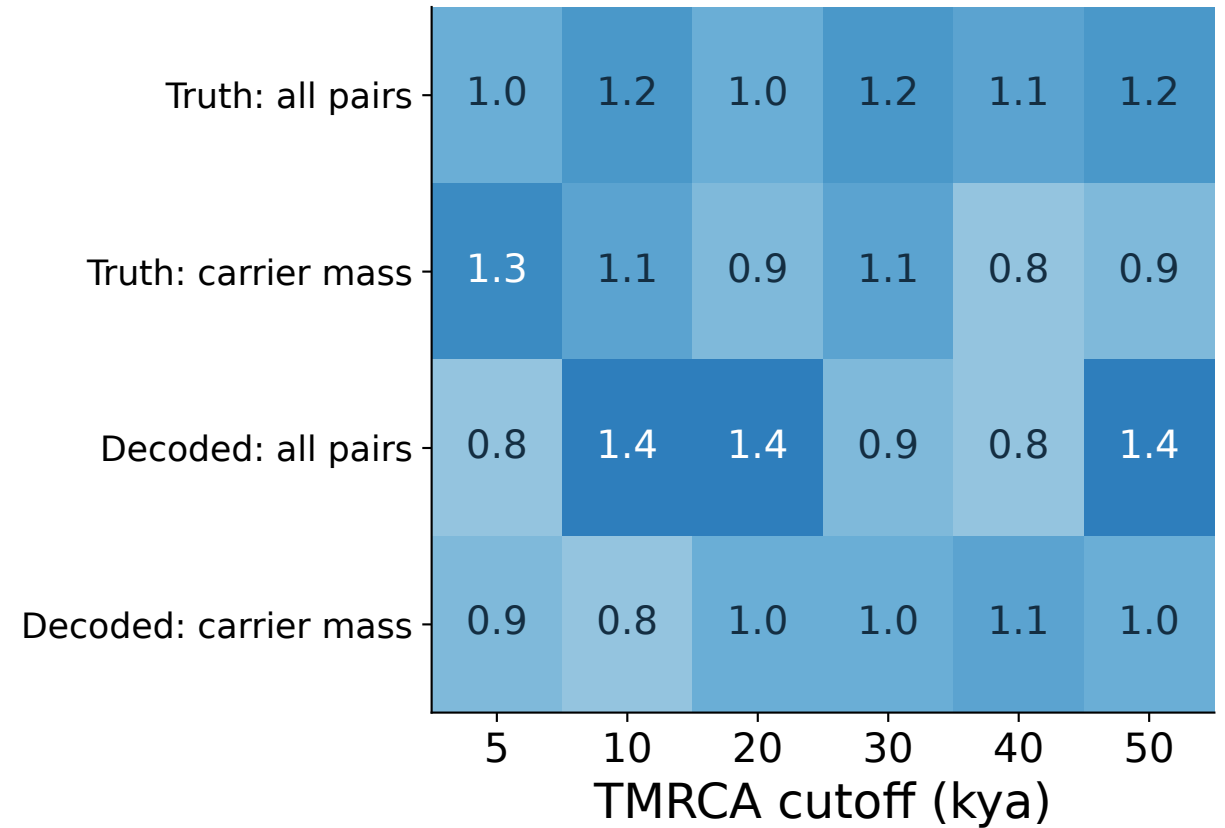
Held-out positional false-positive rates

1,000 neutral positions, tested once each | these are FPRs, not FDRs among discoveries

Nominal 5%



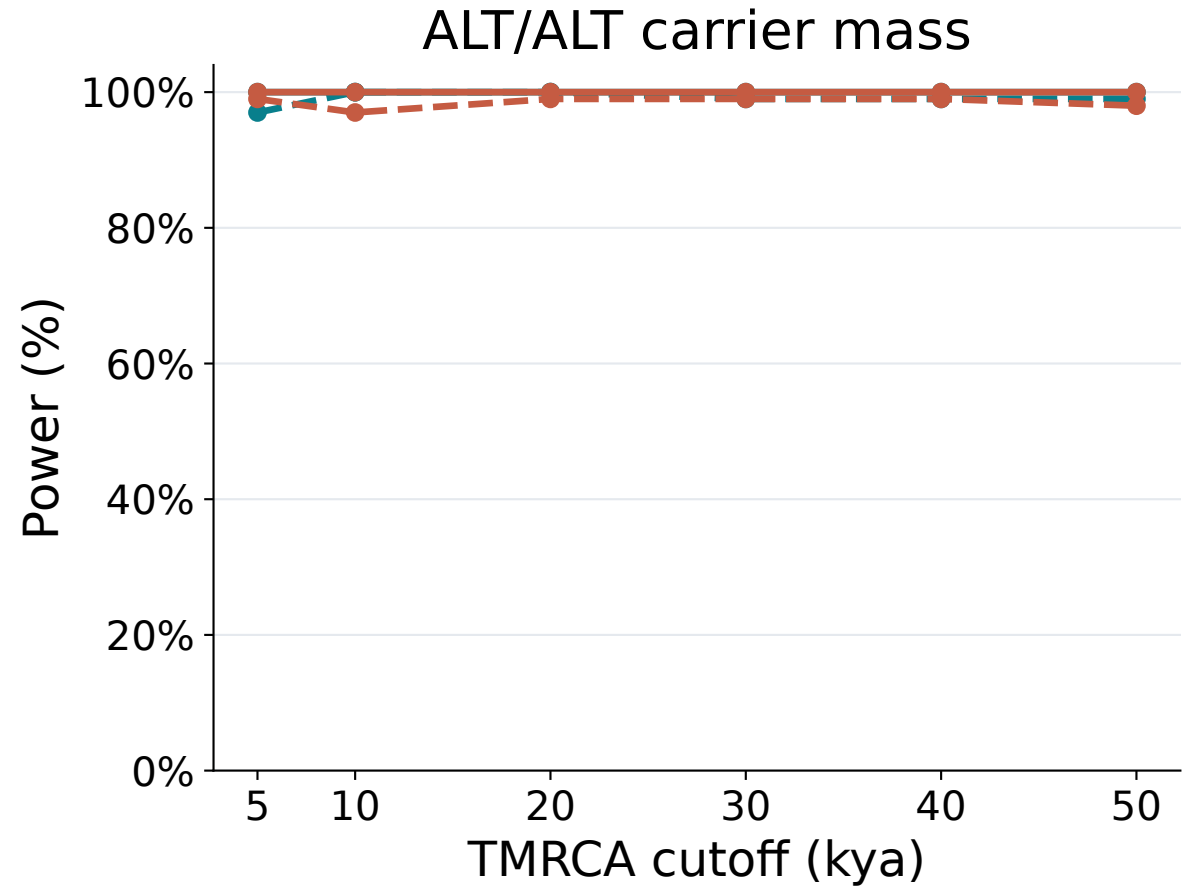
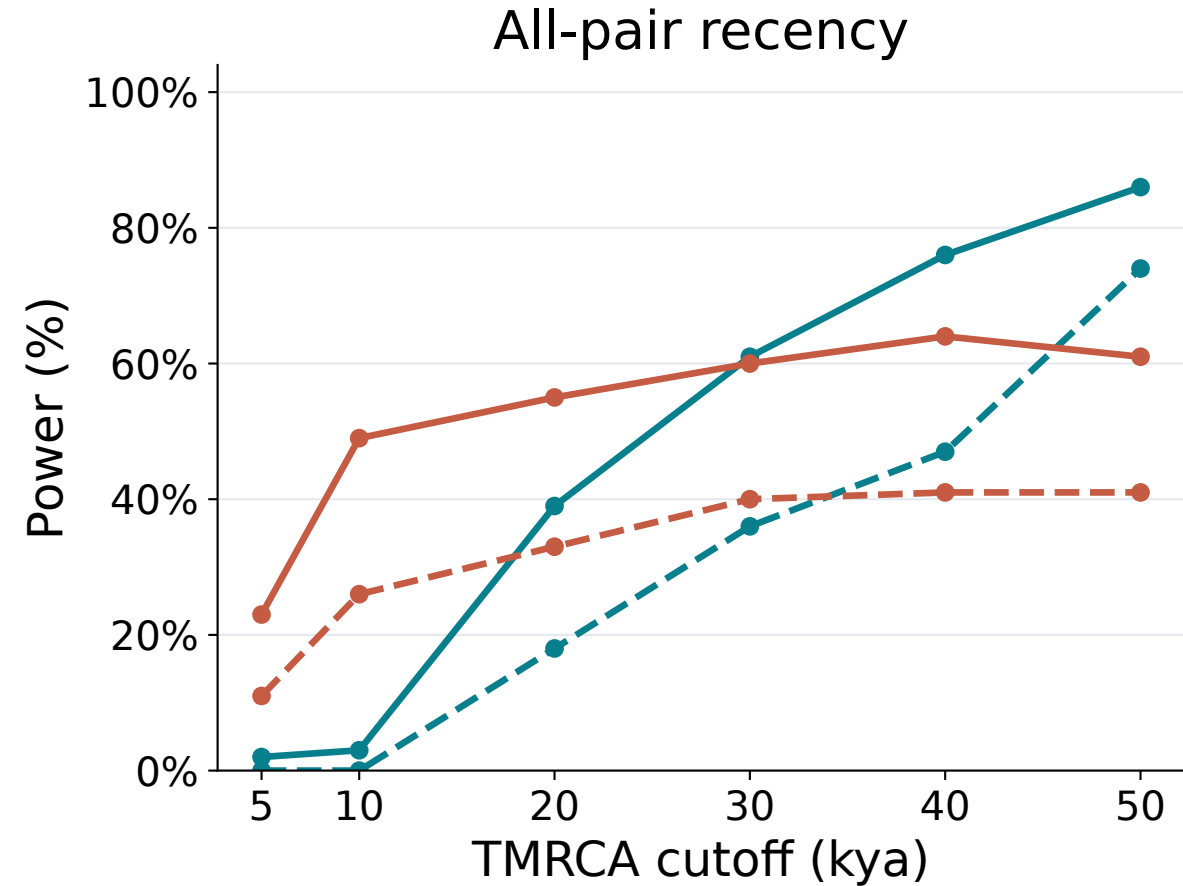
Nominal 1%



Carrier mass retains power after decoding

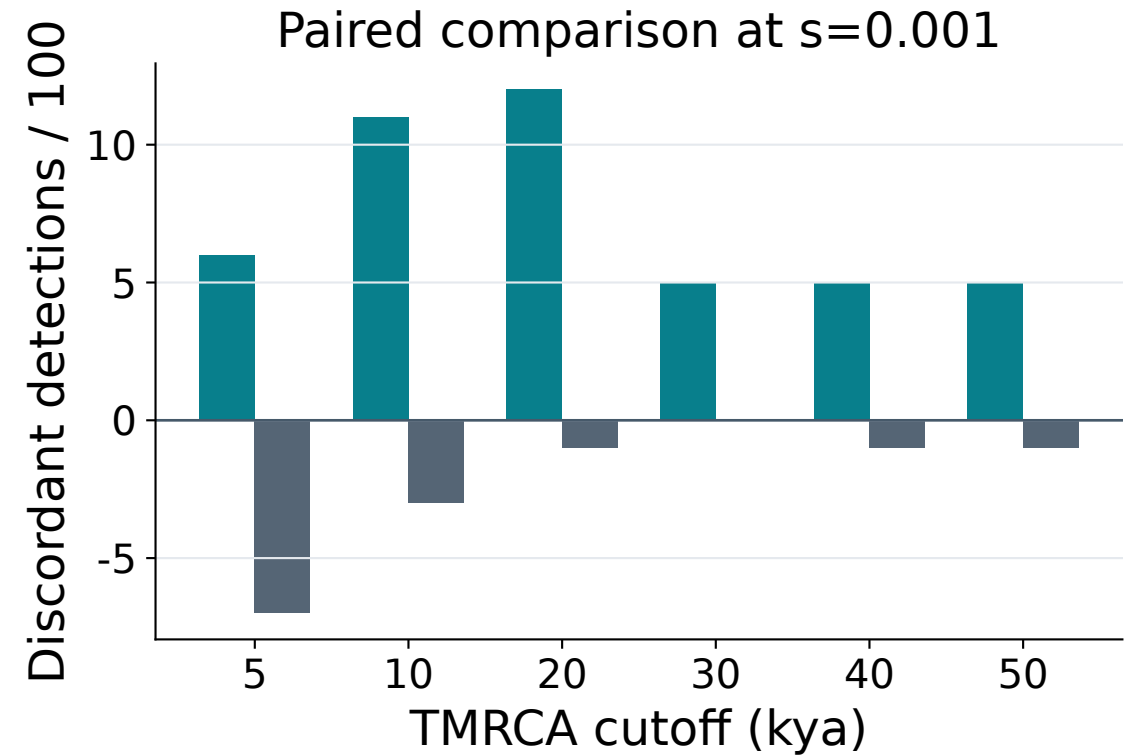
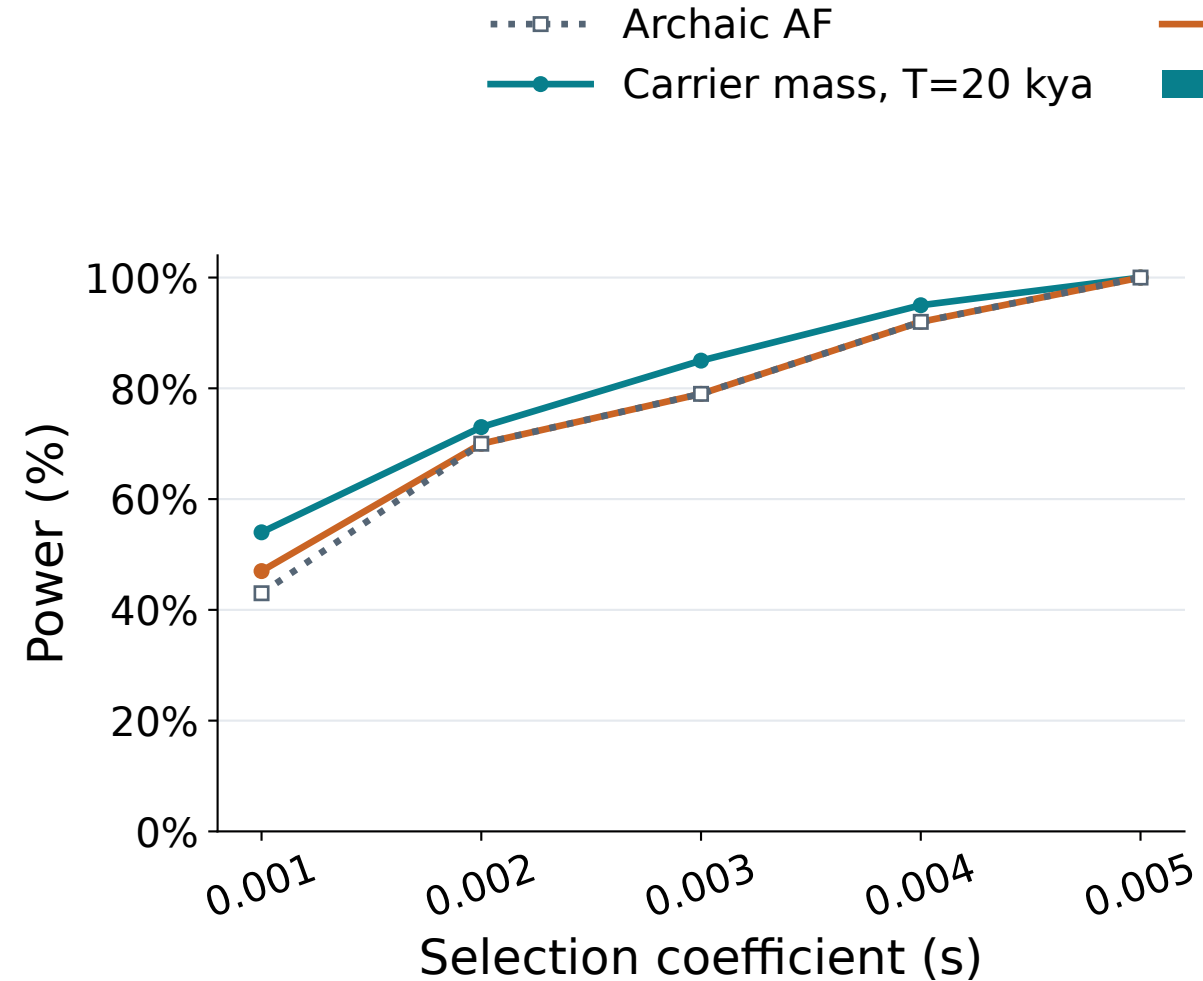
$s = 0.005$ only | matched truth and decoded nulls | 100 selected + 1,000 neutral regions

—●— Truth, $p \leq 0.05$ -●- Truth, $p \leq 0.01$ —●— Decoded, $p \leq 0.05$ -●- Decoded, $p \leq 0.01$



Coalescence can add information beyond AF

True TMRCA | $p \leq 0.05$ | comparisons across T and s are exploratory

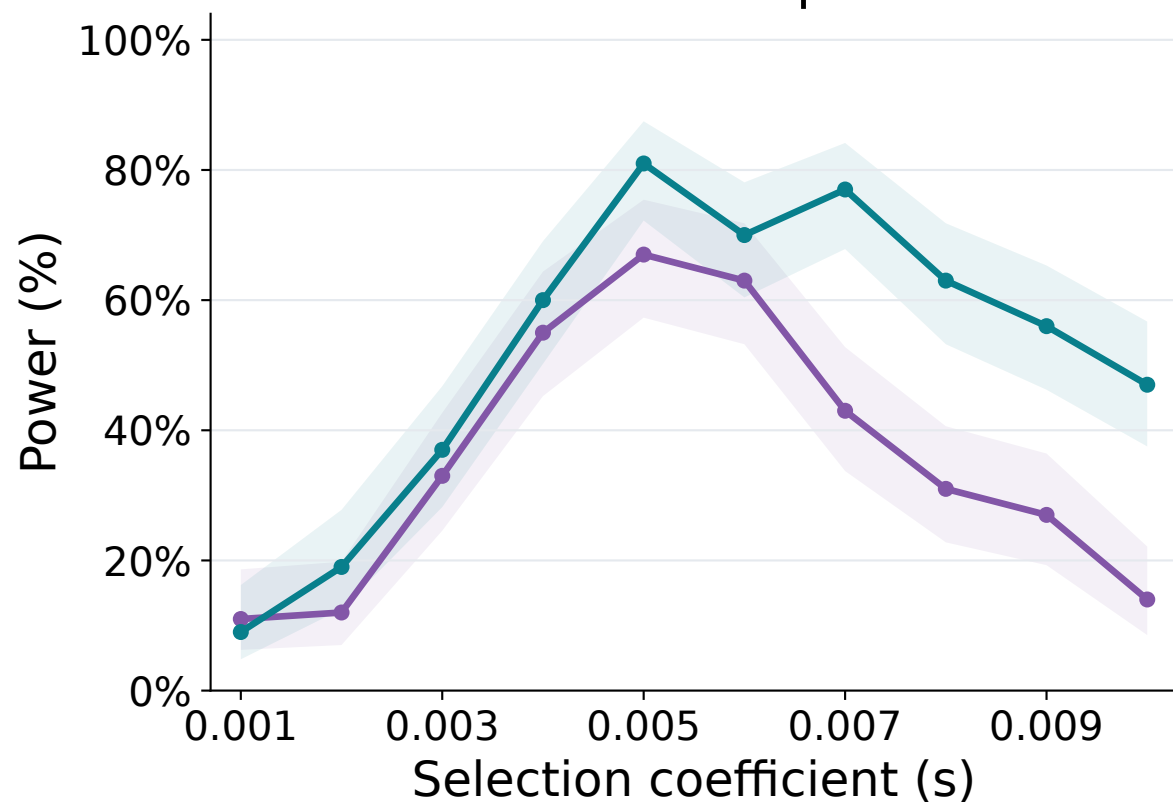


iHS is strongest before the target approaches fixation

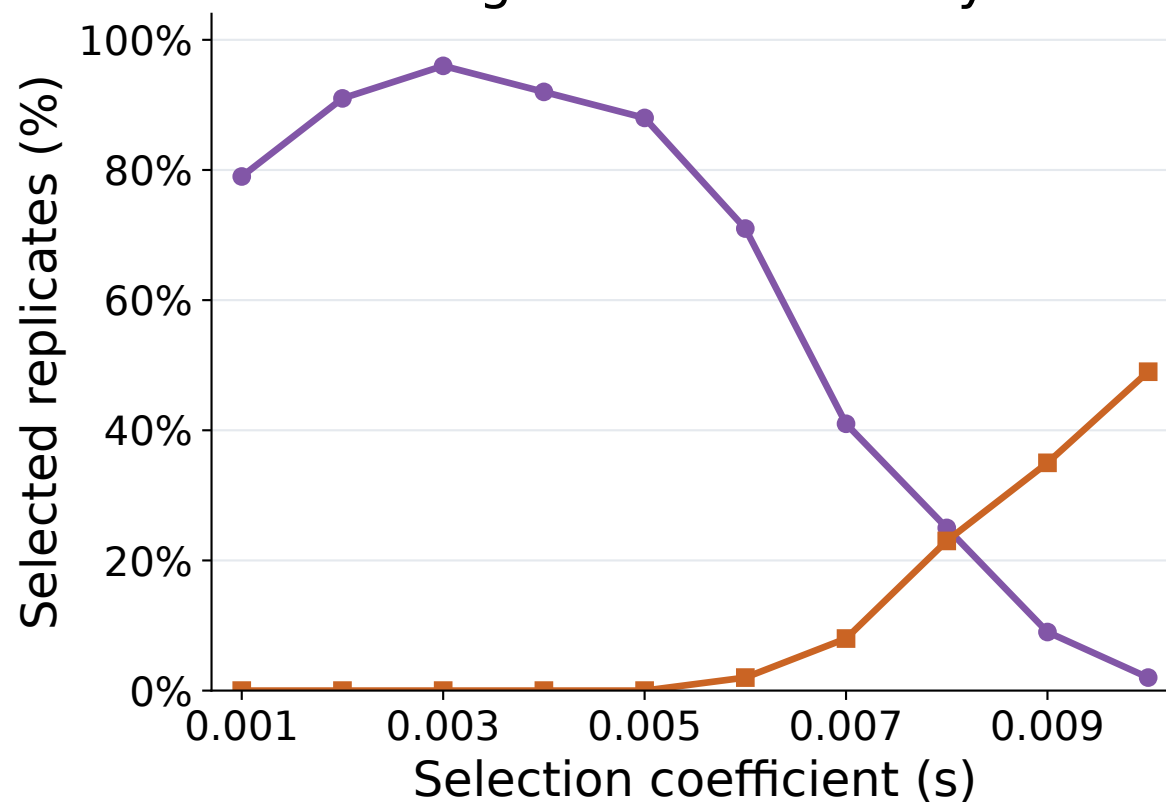
Conventional frequency-standardized iHS | all selected replicates retained

- Positional iHS
- Fixed 100-kb window
- Focal SNP has finite iHS
- Selected allele fixed

Power at nominal $p \leq 0.05$



Target-site availability

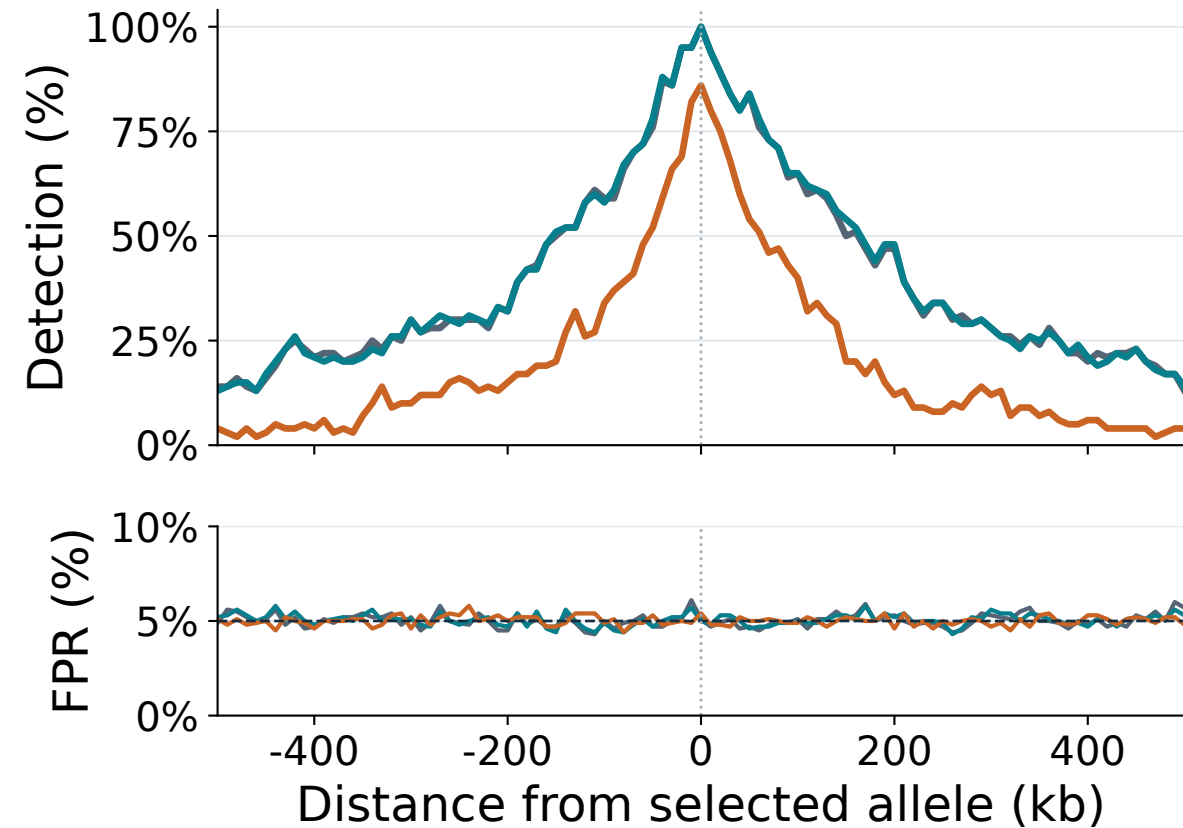


The selected signal decays with distance

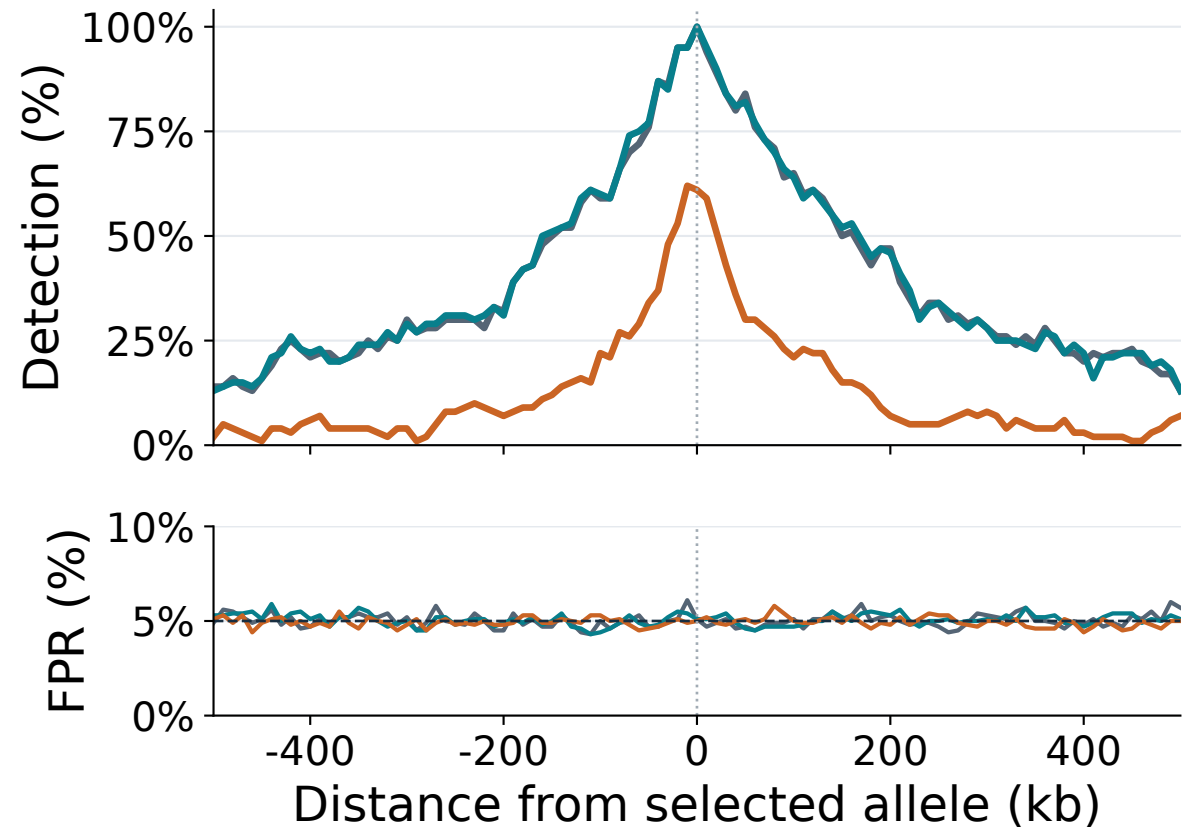
$s = 0.005$; $T = 50$ kya; $p \leq 0.05$ | every stride uses its own matched neutral position

— Archaic AF — Carrier mass — All-pair recency

Truth



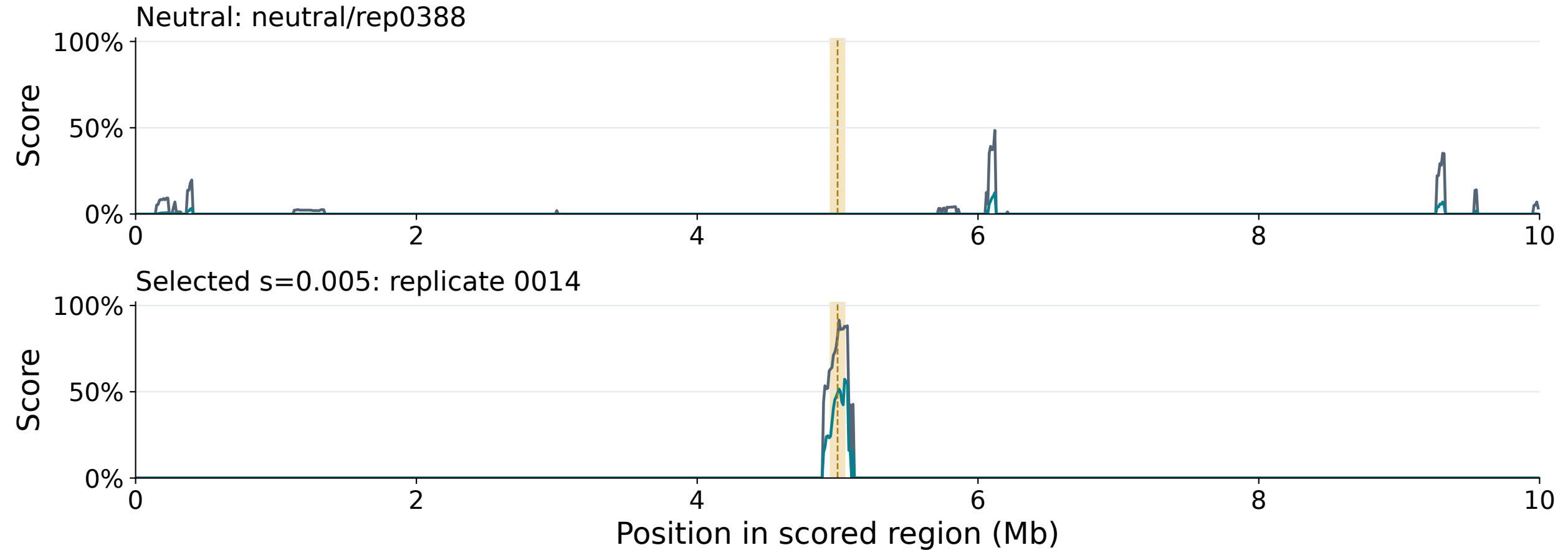
Decoded



A peak elsewhere does not invalidate the focal region

Illustrative examples selected by fixed rules; not an estimate of power

— Archaic AF — Decoded carrier mass

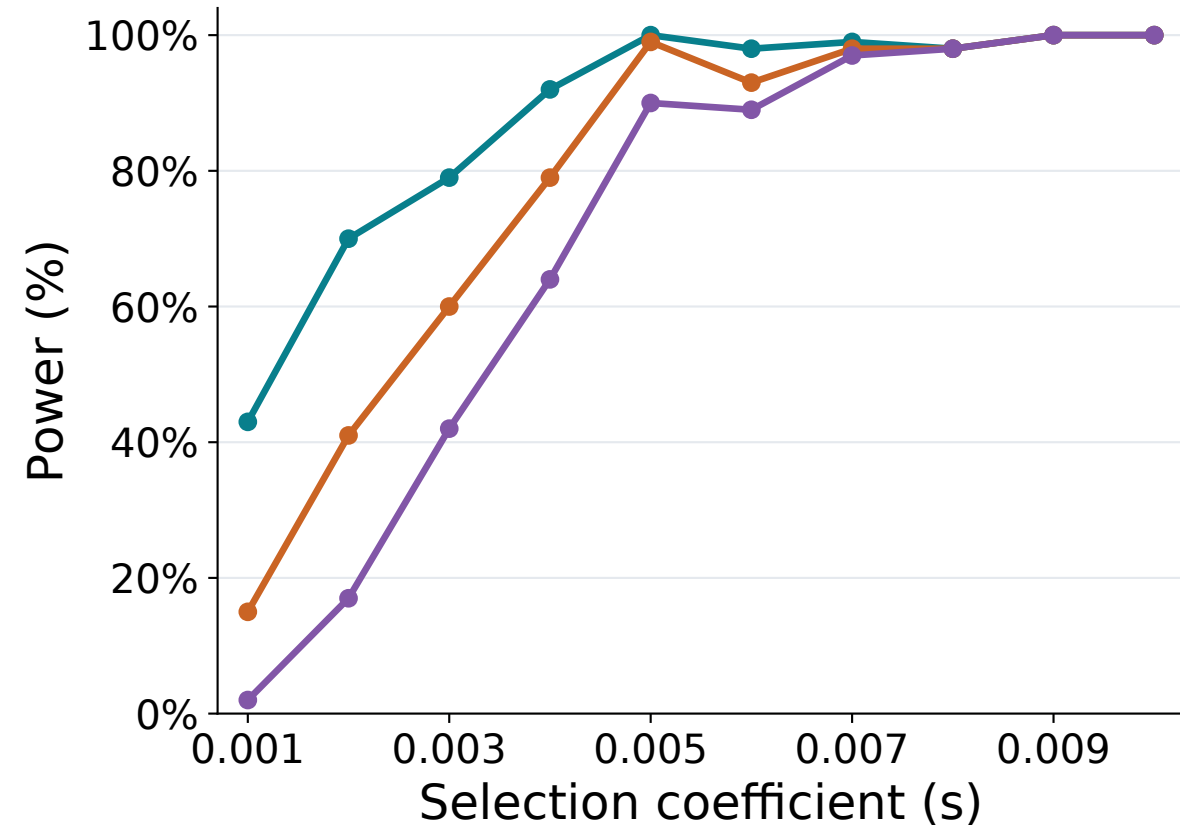


Greater stringency trades power for fewer false calls

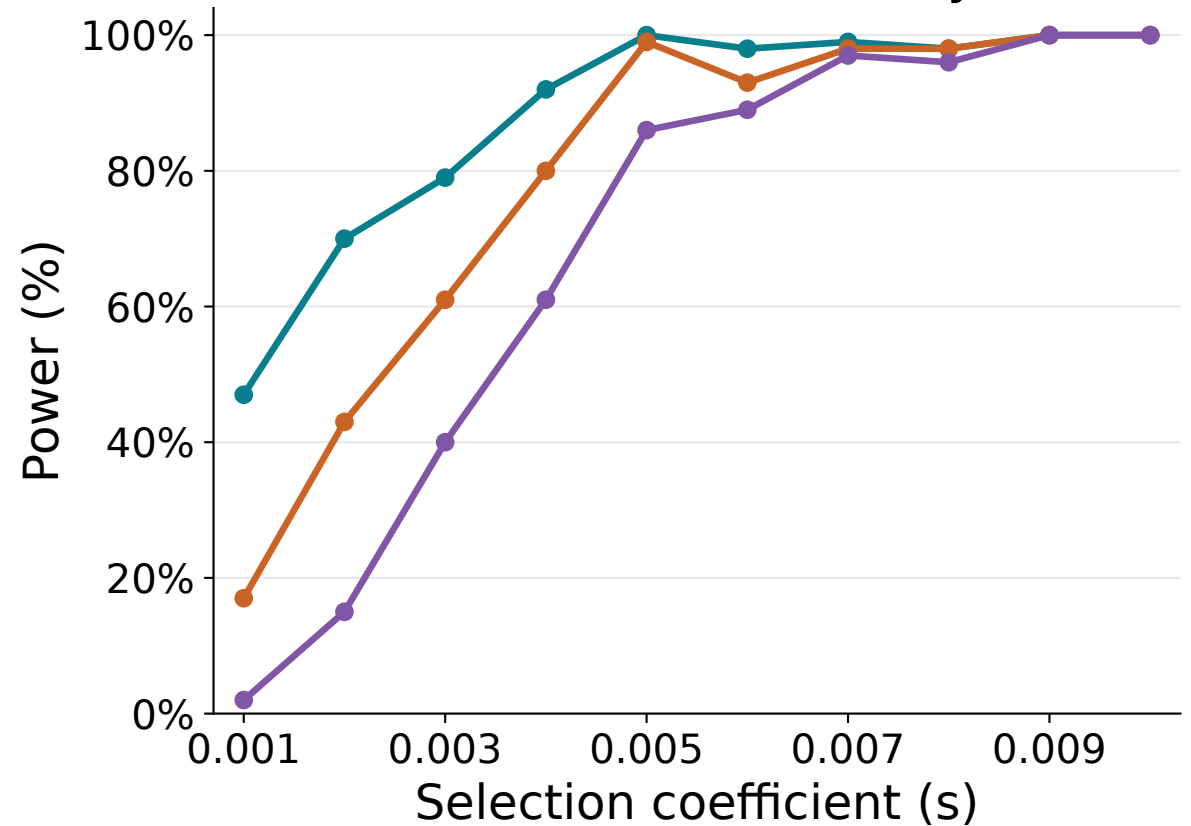
Pointwise tests | $p=0.001$ uses pooled ranks and has very limited neutral-tail resolution

—●— $p \leq 0.05$ (5-fold) —●— $p \leq 0.01$ (5-fold) —●— $p \leq 0.001$ (pooled)

Archaic AF



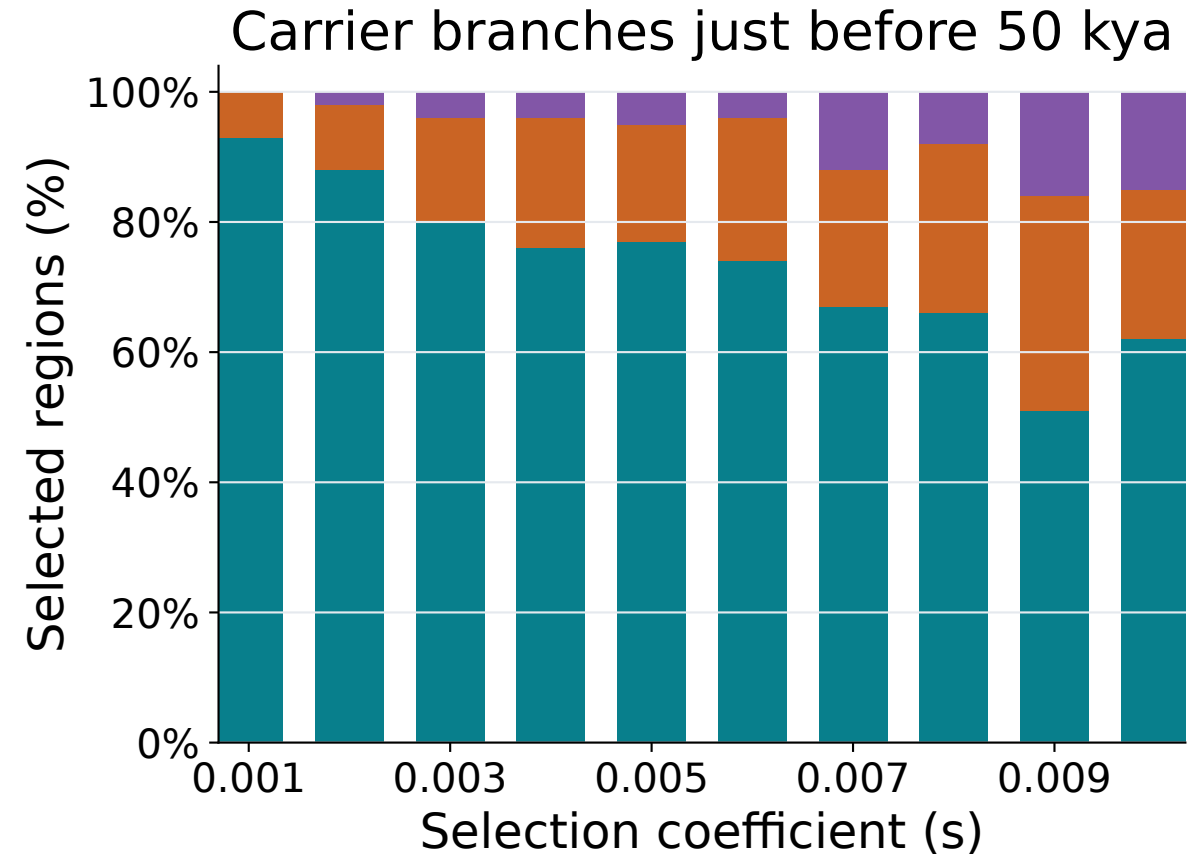
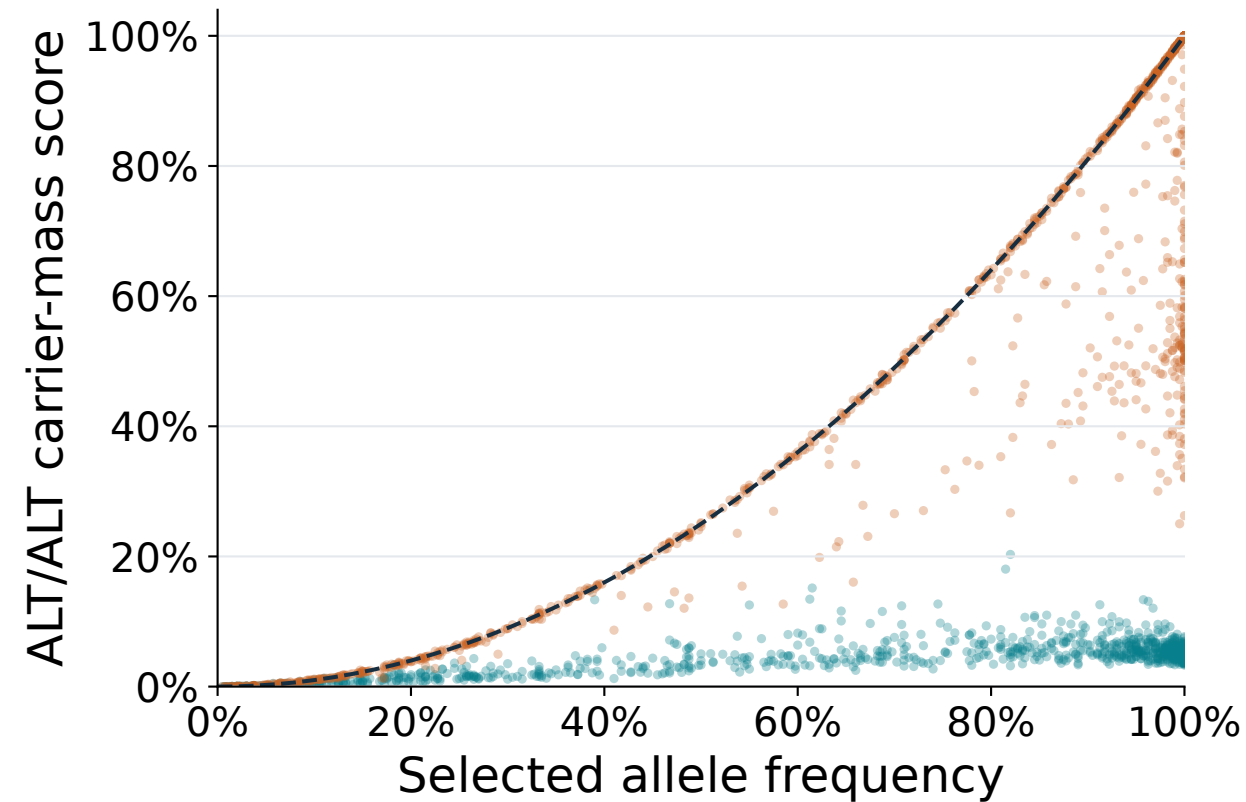
Carrier mass, T=50 kya



Carrier mass and allele frequency are related

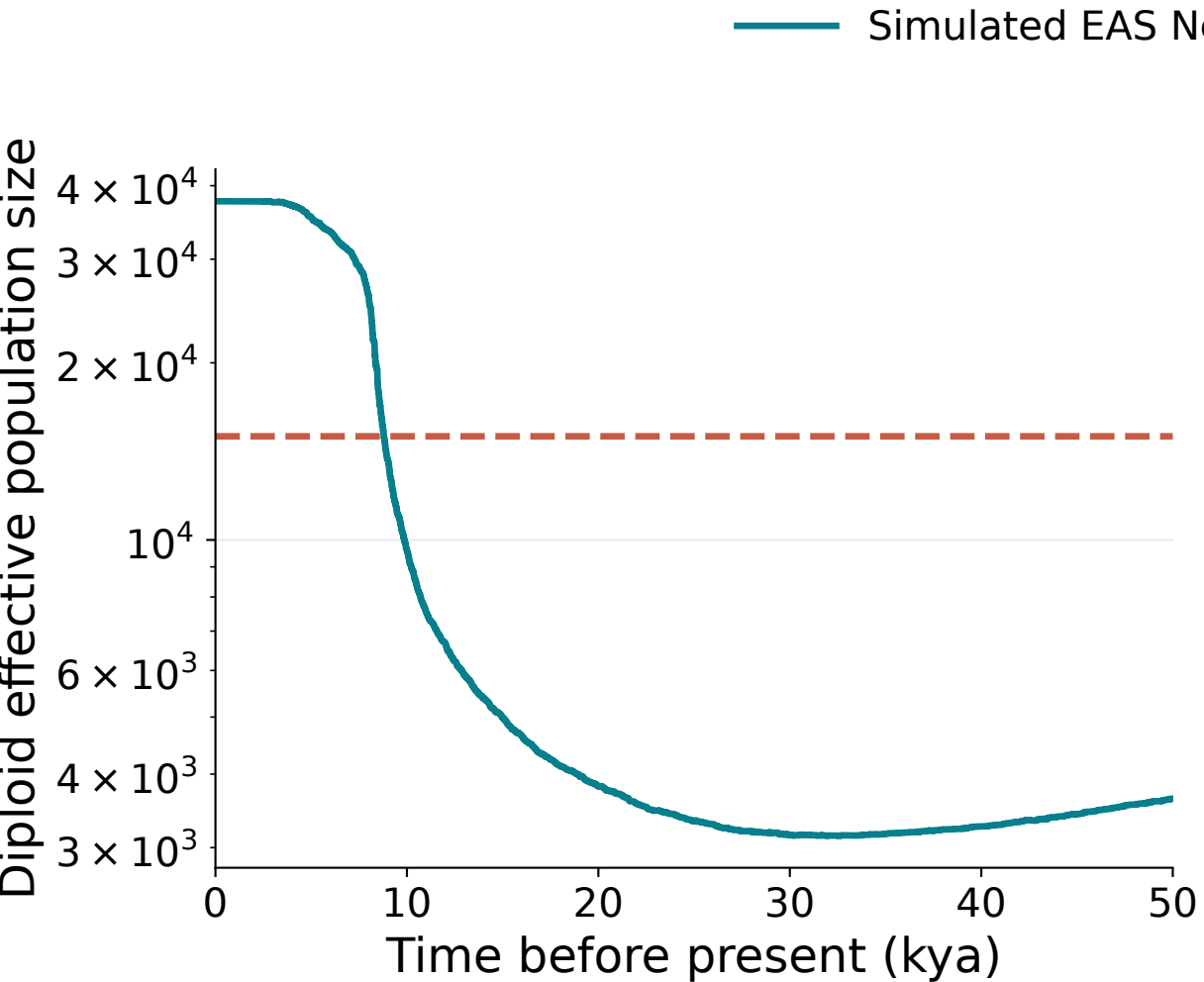
True focal genealogies | 1,000 selected regions | ancestral branches are not a count of founding haplotypes

- T=20 kya
- T=50 kya
- AF squared (approximation)
- One branch
- Two branches
- Three or more



Demography is shared by neutral and selected simulations

The generative EAS history varies with time; the current decoder uses a fixed Ne



Archaic branch

Split: 700 kya

Baseline Ne: 3,600

Pulse bottleneck: Ne=10
for 100 generations

Mutation rate: 1.25×10^{-8}

Recombination rate: 1×10^{-8}

Generation time: 25 years

Recent coalescence among archaic ALT/ALT pairs

Focal position | pairs pooled within each defined class | $s=0.005$ selected versus neutral

Selected: ALT/ALT Selected: REF/REF Neutral: ALT/ALT Neutral: REF/REF

